

## DESERT RESORTS, CA, USA

WMO: 747187

Lat: 33.627N

Lon: 116.159W

Elev: -36

StdP: 101.76

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 03104

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|--------------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                          |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                      | (o)  | (p)   |
| 12            | -0.2       | 1.6 | -15.3                         | 1.0 | 18.7 | -12.9 | 1.2 | 18.1 | 9.3                   | 16.7 | 8.1 | 17.0 | 1.4                      | 340  | 0.471 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.3                   | 44.8            | 22.2 | 43.2 | 21.9 | 42.1 | 21.7 | 26.5                | 36.1 | 25.7 | 36.1 | 25.0 | 36.1 | 3.4                  | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 23.8                            | 18.6 | 31.6 | 22.7 | 17.3 | 31.8 | 21.4 | 16.0 | 32.3 | 82.0          | 35.9 | 78.6 | 36.0 | 75.5 | 36.0 | 31.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |      |     |
| (4) 9.2           | 8.2  | 7.2 | DB                         | -4.1 | 48.2               | 2.2 | 1.7                                                | -5.6 | 49.4       | -6.9 | 50.4       | -8.1 | 51.4       | -9.7  | 52.7 | (4) |
| (5)               |      |     | WB                         | -6.0 | 27.7               | 2.0 | 1.0                                                | -7.4 | 28.4       | -8.6 | 29.0       | -9.7 | 29.6       | -11.1 | 30.4 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 22.9   | 13.2  | 14.8  | 18.9  | 22.0  | 26.3  | 30.4  | 33.5  | 33.1  | 29.8  | 23.3  | 16.8  | 12.2  | (6)  |      |
|                                                                        | DBStd       | 8.08   | 3.09  | 3.29  | 3.55  | 3.64  | 3.39  | 3.10  | 2.51  | 2.48  | 2.97  | 3.33  | 3.59  | 2.99  | (7)  |      |
|                                                                        | HDD10.0     | 24     | 8     | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 12    | (8)  |      |
|                                                                        | HDD18.3     | 577    | 162   | 104   | 37    | 9     | 1     | 0     | 0     | 0     | 0     | 4     | 69    | 190   | (9)  |      |
|                                                                        | CDD10.0     | 4738   | 106   | 138   | 277   | 360   | 504   | 613   | 730   | 717   | 593   | 414   | 206   | 81    | (10) |      |
|                                                                        | CDD18.3     | 2249   | 2     | 7     | 55    | 119   | 246   | 363   | 472   | 459   | 343   | 160   | 23    | 1     | (11) |      |
|                                                                        | CDH23.3     | 36493  | 111   | 242   | 960   | 1753  | 3492  | 5973  | 8153  | 7832  | 5233  | 2177  | 516   | 52    | (12) |      |
| (13)                                                                   | CDH26.7     | 23712  | 16    | 64    | 412   | 895   | 2092  | 4068  | 5821  | 5523  | 3437  | 1198  | 183   | 5     | (13) |      |
| (14)                                                                   | Wind        | WSAvg  | 2.9   | 2.1   | 2.5   | 3.2   | 3.9   | 4.0   | 3.6   | 3.1   | 2.9   | 2.8   | 2.6   | 2.1   | 2.0  | (14) |
| (15) Precipitation                                                     | PrecAvg     | 64     | 9     | 15    | 9     | 1     | 2     | 1     | 2     | 6     | 6     | 2     | 6     | 5     | (15) |      |
|                                                                        | PrecMax     | 167    | 62    | 86    | 43    | 13    | 20    | 10    | 27    | 92    | 60    | 13    | 43    | 25    | (16) |      |
|                                                                        | PrecMin     | 24     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (17) |      |
|                                                                        | PrecStd     | 44     | 16    | 21    | 11    | 3     | 4     | 2     | 6     | 17    | 13    | 3     | 10    | 8     | (18) |      |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 28.6  | 31.2  | 35.7  | 39.0  | 42.5  | 46.3  | 47.3  | 46.3  | 44.1  | 39.9  | 33.4  | 27.2  | (19) |      |
|                                                                        |             | MCWB   | 13.2  | 14.6  | 16.2  | 17.8  | 20.5  | 21.6  | 23.8  | 22.5  | 21.8  | 19.5  | 16.0  | 13.7  | (20) |      |
|                                                                        | 2%          | DB     | 26.2  | 28.7  | 33.1  | 36.4  | 39.9  | 43.5  | 44.7  | 44.4  | 42.2  | 37.3  | 31.1  | 24.7  | (21) |      |
|                                                                        |             | MCWB   | 12.8  | 13.7  | 15.5  | 16.8  | 19.0  | 20.6  | 22.6  | 22.6  | 21.4  | 18.7  | 15.5  | 12.8  | (22) |      |
|                                                                        | 5%          | DB     | 24.1  | 26.5  | 31.2  | 34.0  | 37.8  | 41.9  | 43.0  | 42.8  | 40.7  | 35.1  | 28.8  | 22.7  | (23) |      |
|                                                                        |             | MCWB   | 12.0  | 13.2  | 14.9  | 16.1  | 18.5  | 20.4  | 22.2  | 22.5  | 21.1  | 17.6  | 14.7  | 11.7  | (24) |      |
|                                                                        | 10%         | DB     | 22.2  | 24.0  | 28.8  | 31.9  | 36.0  | 40.2  | 41.8  | 41.3  | 39.0  | 33.1  | 26.4  | 20.9  | (25) |      |
|                                                                        |             | MCWB   | 11.4  | 12.3  | 14.1  | 15.2  | 17.9  | 19.9  | 22.1  | 22.3  | 20.9  | 17.2  | 13.9  | 10.9  | (26) |      |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 15.7  | 16.8  | 17.6  | 19.4  | 22.7  | 25.0  | 27.0  | 27.7  | 26.9  | 22.7  | 18.0  | 15.9  | (27) |      |
|                                                                        |             | MCDB   | 23.5  | 26.3  | 30.8  | 35.8  | 39.2  | 39.6  | 37.2  | 36.6  | 35.1  | 32.4  | 28.5  | 23.8  | (28) |      |
|                                                                        | 2%          | WB     | 14.4  | 15.3  | 16.6  | 17.9  | 20.8  | 23.4  | 26.1  | 26.8  | 25.5  | 21.0  | 16.6  | 14.4  | (29) |      |
|                                                                        |             | MCDB   | 21.7  | 24.0  | 29.8  | 33.2  | 35.7  | 37.7  | 36.5  | 36.2  | 35.1  | 31.7  | 28.0  | 21.2  | (30) |      |
|                                                                        | 5%          | WB     | 13.4  | 14.2  | 15.8  | 16.9  | 19.7  | 22.3  | 25.4  | 25.9  | 24.5  | 19.7  | 15.6  | 13.3  | (31) |      |
|                                                                        |             | MCDB   | 21.1  | 23.3  | 28.5  | 31.4  | 34.1  | 37.4  | 36.5  | 36.0  | 34.6  | 32.2  | 26.8  | 20.0  | (32) |      |
|                                                                        | 10%         | WB     | 12.3  | 13.3  | 15.0  | 16.0  | 18.7  | 21.3  | 24.7  | 25.2  | 23.4  | 18.4  | 14.7  | 12.0  | (33) |      |
|                                                                        |             | MCDB   | 20.3  | 22.2  | 26.9  | 29.8  | 33.5  | 36.7  | 36.5  | 36.0  | 34.2  | 30.5  | 25.0  | 18.8  | (34) |      |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 17.2  | 16.9  | 17.4  | 17.0  | 16.7  | 18.3  | 16.3  | 16.4  | 17.6  | 18.1  | 18.2  | 17.0  | (35) |      |
|                                                                        |             | MCDBR  | 20.4  | 20.8  | 21.2  | 21.2  | 19.8  | 19.9  | 17.7  | 17.9  | 19.4  | 21.5  | 21.2  | 19.5  | (36) |      |
|                                                                        |             | MCWBR  | 10.1  | 9.3   | 8.2   | 7.6   | 6.3   | 6.6   | 6.3   | 6.4   | 6.6   | 8.1   | 9.4   | 10.1  | (37) |      |
|                                                                        | 5% WB       | MCDBR  | 17.1  | 17.2  | 18.9  | 19.1  | 18.8  | 19.2  | 14.6  | 14.5  | 15.3  | 18.4  | 18.7  | 16.1  | (38) |      |
|                                                                        |             | MCWBR  | 9.0   | 8.3   | 7.6   | 7.1   | 6.4   | 7.3   | 6.2   | 6.8   | 7.1   | 7.7   | 8.7   | 8.9   | (39) |      |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.305 | 0.308 | 0.328 | 0.352 | 0.369 | 0.369 | 0.405 | 0.395 | 0.360 | 0.337 | 0.309 | 0.308 | (40) |      |
|                                                                        | taud        |        | 2.496 | 2.505 | 2.435 | 2.373 | 2.353 | 2.368 | 2.367 | 2.380 | 2.446 | 2.470 | 2.524 | 2.521 | (41) |      |
|                                                                        | Ebn at Noon |        | 912   | 947   | 949   | 934   | 916   | 911   | 877   | 882   | 903   | 901   | 901   | 888   | (42) |      |
|                                                                        | Edh at Noon |        | 85    | 94    | 109   | 121   | 125   | 123   | 122   | 119   | 106   | 96    | 82    | 79    | (43) |      |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 3.35  | 4.30  | 5.79  | 7.08  | 7.96  | 8.45  | 7.70  | 7.18  | 6.26  | 4.93  | 3.71  | 3.03  | (44) |      |
|                                                                        | RadStd      |        | 0.21  | 0.36  | 0.26  | 0.22  | 0.28  | 0.30  | 0.34  | 0.30  | 0.23  | 0.24  | 0.16  | 0.18  | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |      |  |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|------|--|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |      |  |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |      |  |
| (46) | Station Only            | +0.66 | N/A     | -1.69  | N/A     | -0.42 | N/A   | N/A         | N/A     | +215    | +156    | (46) |  |
| (47) | Regional (38 neighbors) | +0.54 | N/A     | -1.65  | +0.59   | N/A   | N/A   | N/A         | -83     | +192    | +104    | (47) |  |

Nomenclature: See separate page